

High-Quality Perception Data

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Energy Efficient and Autonomous Vehicles (EEAV) Lab

Mechanical and Aerospace Engineering Department

Western Michigan University

June 5th, 2024

Project ID: EEMS116

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Overview

Timeline	Barriers and Technical Targets
• Project Start Date: Oct 2021 • Project End Date: Dec 2024 • Percent Complete: 100%	Vehicle-Mobility Systems Analysis Tech Team (VMSATT) Roadmap 1. Evaluation of future mobility scenarios (on-board sensing & computing) 2. Evaluation of Vehicle Technical Specifications (VTS) and Operational Design Domain (ODD) 3. Inventory performance and develop improvements in on-board sensing & computing 4. Identify synergies between components and new mobility technologies 5. Engage with external stakeholders by sharing research outcomes
Budget	**Partners**
• Total Project Funding: $2.5m ○ DOE share: $2m ○ Contractor share: $0.5m • Funding for FY 2022: $1.1m • Funding for FY 2023: $0.75m	**Oak Ridge National Laboratory** • Ali Riza Ekti, PhD • Ross Wang, PhD **National Center for Atmospheric Research** • Curtis Walker, PhD • Amanda Siems-Anderson **DriveU.auto** • Eli Shapira • Doron Elinav

Relevance

Objective

- Integrate all energy efficiency enabling sensors with an electric vehicle platform
- Measure an equivalent vehicle operational performance of these new sensors vs. traditional sensors
- Evaluate the energy savings from a reduced use of traditional sensors and high compute algorithms

Infrastructure-Based Sensor Technology	Current	Project End	Anticipated Energy Efficiency Improvement Mechanism(s)
Chip-Enabled Raised Pavement Markers (CERPMs) 	TRL 2	TRL 5	• Elimination or reduced use of on-board sensors • Reduction in overall compute load needs
Radar Retroreflectors (RRs) 	TRL 2	TRL 5	
Cellular Network Offloading 	TRL 3	TRL 4	• Reduction in overall compute load needs
High Definition (HD) Maps 	TRL 5	TRL 6	• Reduction in overall compute load needs
Weather Sensors 	TRL 5	TRL 6	• Reduction in overall compute load needs

Impact

- Promotes U.S. energy security by reducing energy use from driving automation technologies
- Energy Efficient Mobility Systems (EEMS) Goal #2: Innovative technology R&D for energy efficiency
- VMSATT Targets: Evaluation and improvement in on-board sensing & computing systems

Milestones

#	Milestone	Due	Status
M1	Data Agreements, Access, and Collaboration	Dec 21	Complete
M2	Equipment On-Hand	Mar. 22	Complete
M3	HD Map Data and RTK Kit Successful	June 22	Complete
M4	Fusion and Baseline Documentation	Sept. 22	Complete
GNG1	**Viability Demonstration for All Sensors**	**Dec. 22**	**Complete**
M6	Verification of HD Map and RTK Communication with the Vehicle	Mar. 23	Complete
M7	Verification of Communication Between All Sensors and the Vehicle	June 23	Complete
M8	Energy Efficient Fusion Demonstration of Traditional and New Sensors	Sept. 23	Complete
GNG2	**Compute Load Reduction Observed**	**Dec. 23**	**Complete**
M10	On-Road Demonstration Routes Selected	Mar. 24	Complete
M11	On-Road Demonstration of All Developed Sensors	June 24	Complete
M12	Communication to Dept. of Transp., Metro. Plan. Org., & Stakeholders	Sept. 24	Complete
M13	Final Report Delivered to DOE	Dec. 24	Complete

Approach

No Communication



Traditional Auxiliary Power Load

Infrastructure-Based Sensor Technology

- Chip-Enabled Raised Pavement Markers (CERPMs)
- Radar Retroreflectors (RRs)
- Cellular Network Offloading
- High Definition (HD) Maps
- Weather Sensors

VS

High-Quality External Perception Data

- ➔ Fewer On-Board Sensors
- ➔ Less On-Board Computing

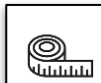


Reduced Auxiliary Power Load

Public domain image, source: https://commons.wikimedia.org/wiki/File:Street_Fighter_VS_logo.png

Approach

	Baseline	Infrastructure-Enabled Perception
Object Detection	• 2D: You Look Only Once applied on images from six cameras • 3D: Complex You Look Only Once applied on LiDAR point cloud	***Reduced compute load as images from 4 cameras were offloaded to the cloud computer for processing***
Object Tracking	Extended Kalman Filter (EKF) to fuse 2D and 3D detections	***Same approach. Turn off underperforming sensors during inclement weather conditions***
Driving Lane Detection	Segnet (encoder decoder architecture)	Kalman filter to fuse CERPMs, radar-retroreflectors, and HD maps. ***Eliminates image processing entirely.***
Localization	Fusion of LiDAR generated odometry using FASTLIO, GNSS (without RTK), and IMU	Fusion of HD Maps, IMU, and GNSS (RTK enabled). ***3D point cloud processing eliminated.***



Traditional Auxiliary Power Load



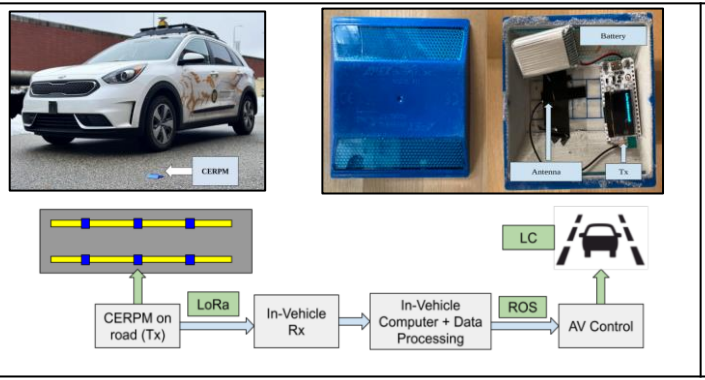


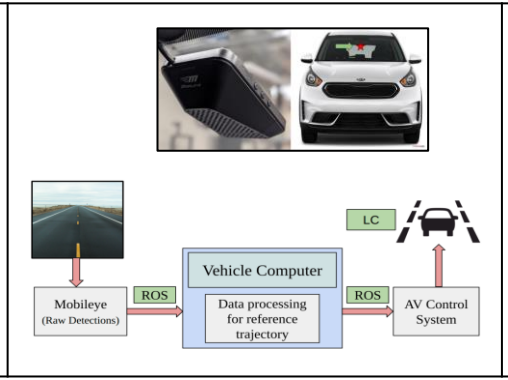
Reduced Auxiliary Power Load

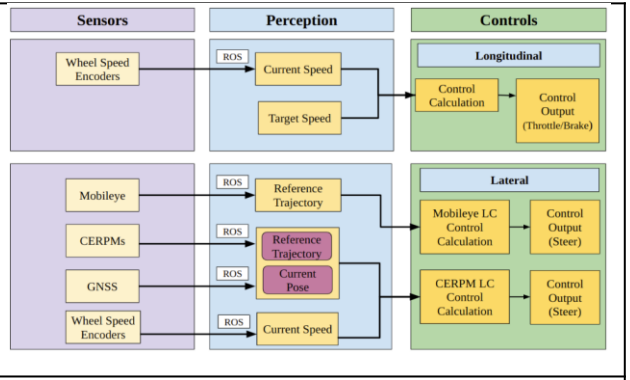
Public domain image, source: https://commons.wikimedia.org/wiki/File:Street_Fighter_VS_logo.png

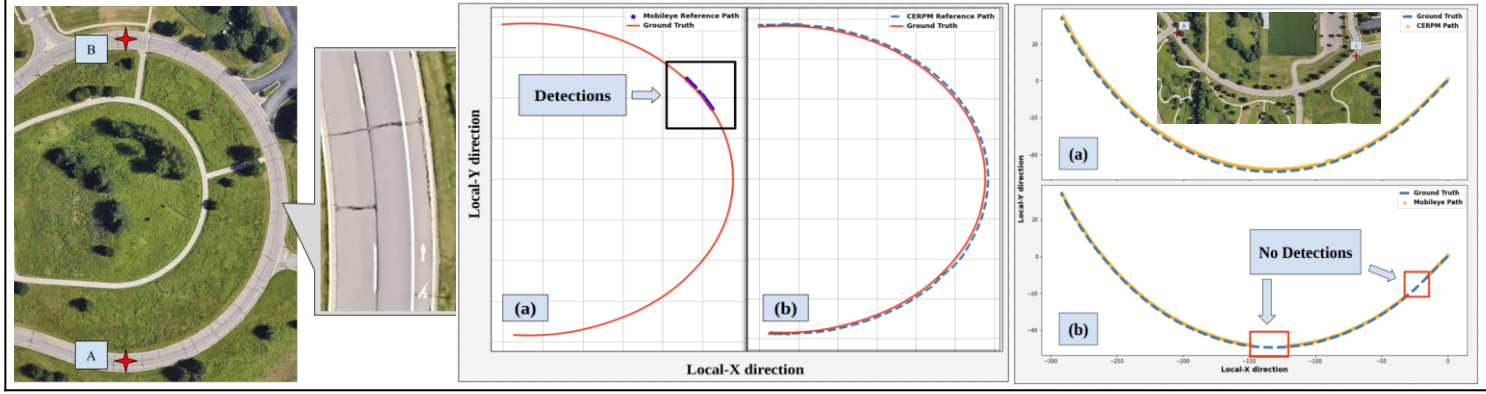
Technical Accomplishments & Progress

Infrastructure Sensor Development: Chip-Enabled Raised Pavement Markers (CERPMs)







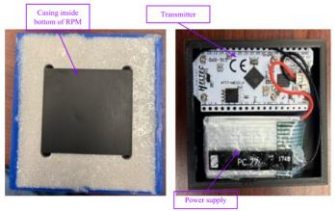


Root-Mean Square of On-Road Vehicle Cross Track Error

CERPM	Mobileye
0.42 m	N/A[1]
0.38 m	0.41 m

"Automated Lane Centering: An Off-the-Shelf Computer Vision Product vs Infrastructure-based Chip-Enabled Raised Pavement Markers" (In Review)
Kadav, Sharma, Fanas, Patil, Wang, Ekti, Meyer, Asher. MDPI Sensors

Technical Accomplishments & Progress



Use existing RPM procedures





CERPMs communicate directly with the vehicle



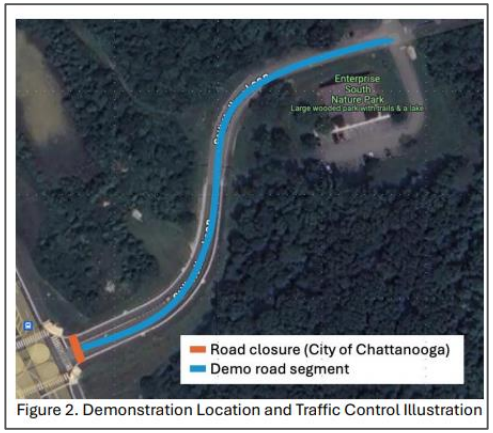
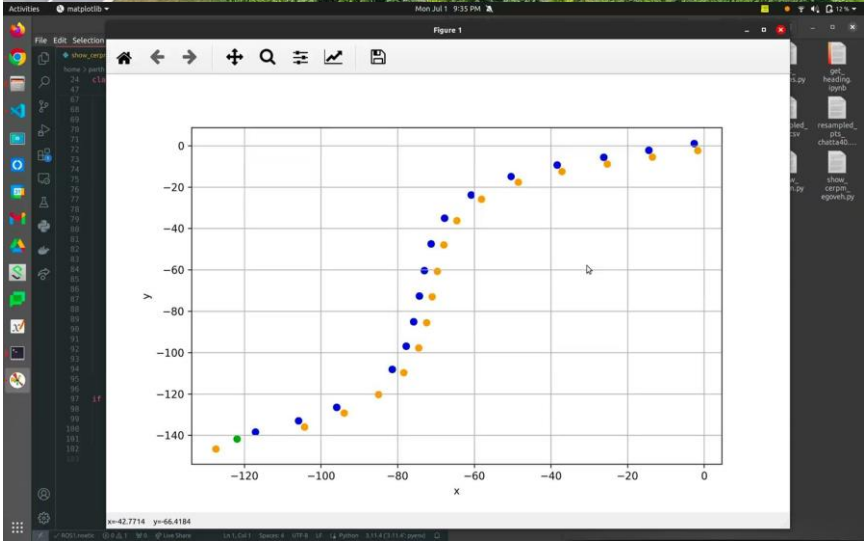


Figure 2. Demonstration Location and Traffic Control Illustration

"Automated Lane Centering: An Off-the-Shelf Computer Vision Product vs Infrastructure-based Chip-Enabled Raised Pavement Markers"
Kadav, Sharma, Fanas, Patil, Wang, Ekti, Meyer, Asher. MDPI Sensors



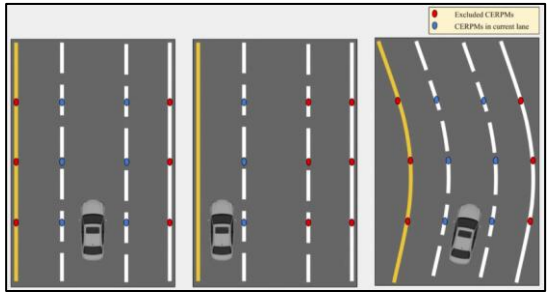

Activities
matplotlib
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Figure 1
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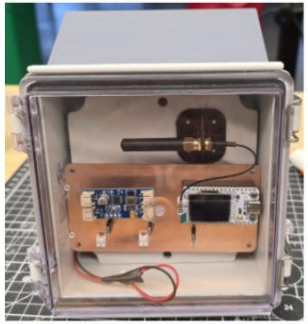




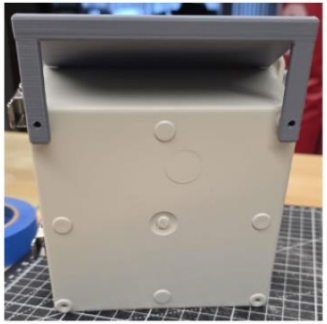
Technical Accomplishments & Progress

CERPM Commercialization (Follow-on DOE SBIR Project)









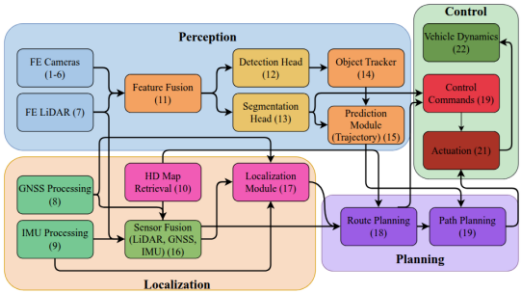


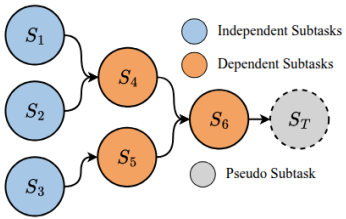


"Automated Lane Centering: An Off-the-Shelf Computer Vision Product vs Infrastructure-based Chip-Enabled Raised Pavement Markers"
Kadav, Sharma, Fanas, Patil, Wang, Ekti, Meyer, Asher. MDPI Sensors

Technical Accomplishments & Progress

Infrastructure Sensor Development: Cellular Network Offloading

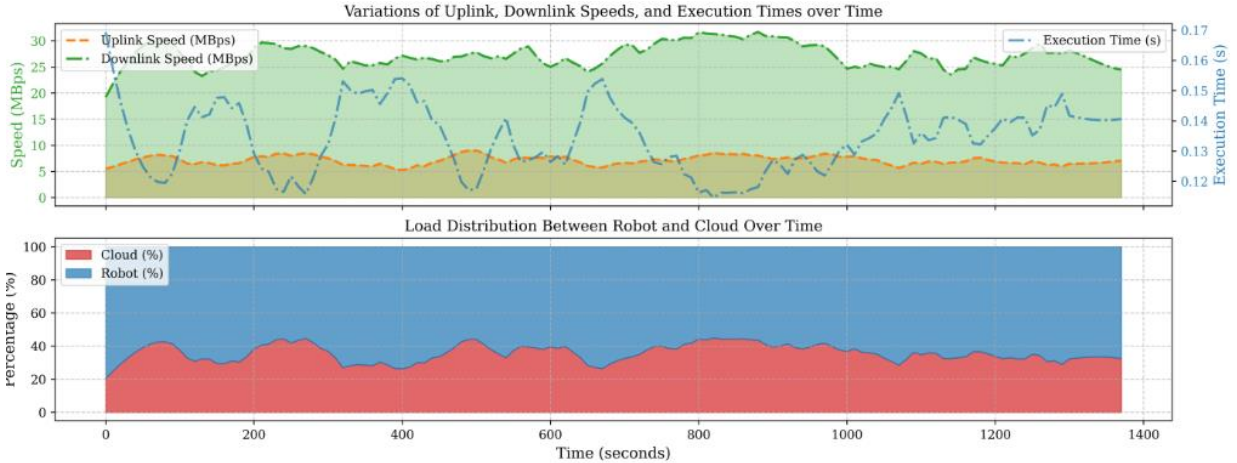




$$\text{Minimize} \quad J_{\text{robot}} = \sum_{i=1}^{n_S} \alpha_{S_i} \Delta t_{S_i}^{M,R}$$

Parameter	Symbol	Value
Robot computational capacity	Y_R	4×10^9 cycles
Cloud computational capacity	Y_C	8×10^9 cycles
Base uplink speed	$\omega_{RC}^{\text{base}}$	8 Mbps
Base downlink speed	$\omega_{CR}^{\text{base}}$	30 Mbps
Standard deviation for uplink speed	σ_{uplink}	2 Mbps
Standard deviation for downlink speed	σ_{downlink}	4 Mbps
Uplink efficiency factor	η_U	1
Downlink efficiency factor	η_D	1

Parameter	Value
Idle Power (Robot)	300 W
Robot CPU Energy per Cycle ($\eta_{\text{CPU}}^{\text{robot}}$)	10 nJoule/cycle
Robot Memory Energy per Cycle ($\eta_{\text{memory}}^{\text{robot}}$)	10 nJoule/cycle
Cloud CPU Energy per Cycle ($\eta_{\text{CPU}}^{\text{cloud}}$)	5 nJoule/cycle
Cloud Memory Energy per Cycle ($\eta_{\text{memory}}^{\text{cloud}}$)	5 nJoule/cycle
Uplink Energy per Bit ($E_{\text{bit}}^{\text{up}}$)	78 nJoule/bit
Downlink Energy per Bit ($E_{\text{bit}}^{\text{down}}$)	6.7 nJoule/bit
Robot CPU Frequency (f_{CPU})	4 GHz

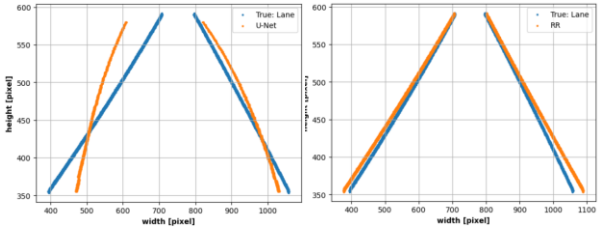


"Latency and Energy Aware Optimal Scheduling Scheme for Computational Offloading of Autonomous Vehicle Perception Data (TBD)
Sharma, Meyer, Asher

Technical Accomplishments & Progress

Infrastructure Sensor Development: Radar Retroreflectors (RRs)



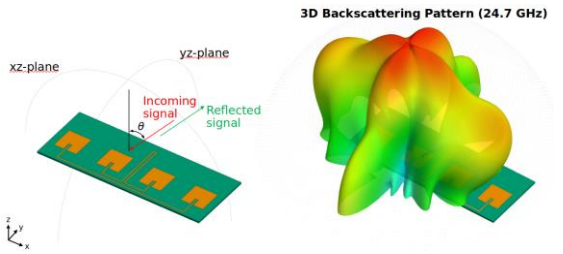


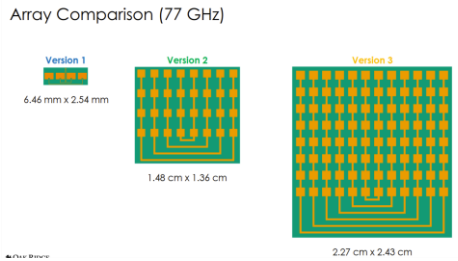


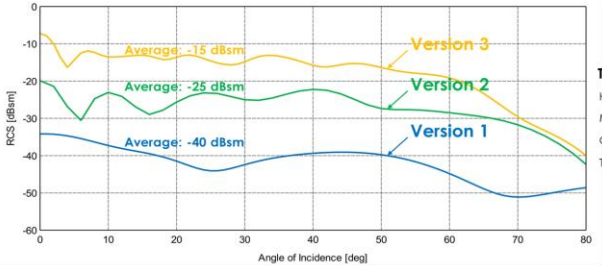


MSE [pixels]	Left Lane Line	Right Lane Line
U-Net	83	73
Radar RRs	5	6









Typical RCS (77 GHz)

Human	-10 dBsm
Motorcycle	7 dBsm
Car (Sedan)	20 dBsm
Trihedral (9 in)	29 dBsm (max)

*Note: dBsm = dB square meter

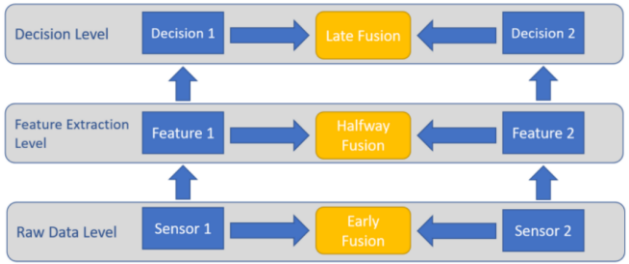
"Real World Use Case Evaluation of Radar Retro-reflectors for Autonomous Vehicle Lane Detection Applications" (2024)
Brown, Patil, Kadav, Sharma, Fanas, Meyer, Hong, Liao, Ekti, Wang, Asher. SAE World Congress

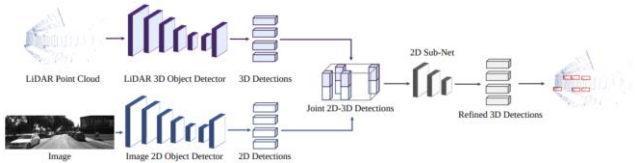
Technical Accomplishments & Progress

Energy Efficient Sensor Fusion

	Traditional Sensor Fusion Approach	New Energy Efficient Sensor Fusion Approach
Overall sensor suite	***Camera, LiDAR, RADAR, GNSS, wheel odometry***	***Camera, RADAR, LiDAR, GNSS w/ RTK corrections, wheel odometry, CERPMs, RRs, HD maps, weather sensors***
Object perception	Camera, Radar, Lidar: Camera & Lidar You Only Look Once, RANSAC, EKF fusion	Camera, Radar, Lidar: Toggling based on weather conditions
Lane line perception	Camera: U-Net and/or Segnet	CERPMs, RRs, HD maps: EKF fusion
Localization	Lidar, GNSS, IMU: Lidar based odometry generation, EKF Fusion	GNSS with RTK, IMU, wheel odometry: EKF fusion

- ➔ Measure auxiliary power consumption of algorithms
- ➔ New combinations of existing fusion techniques



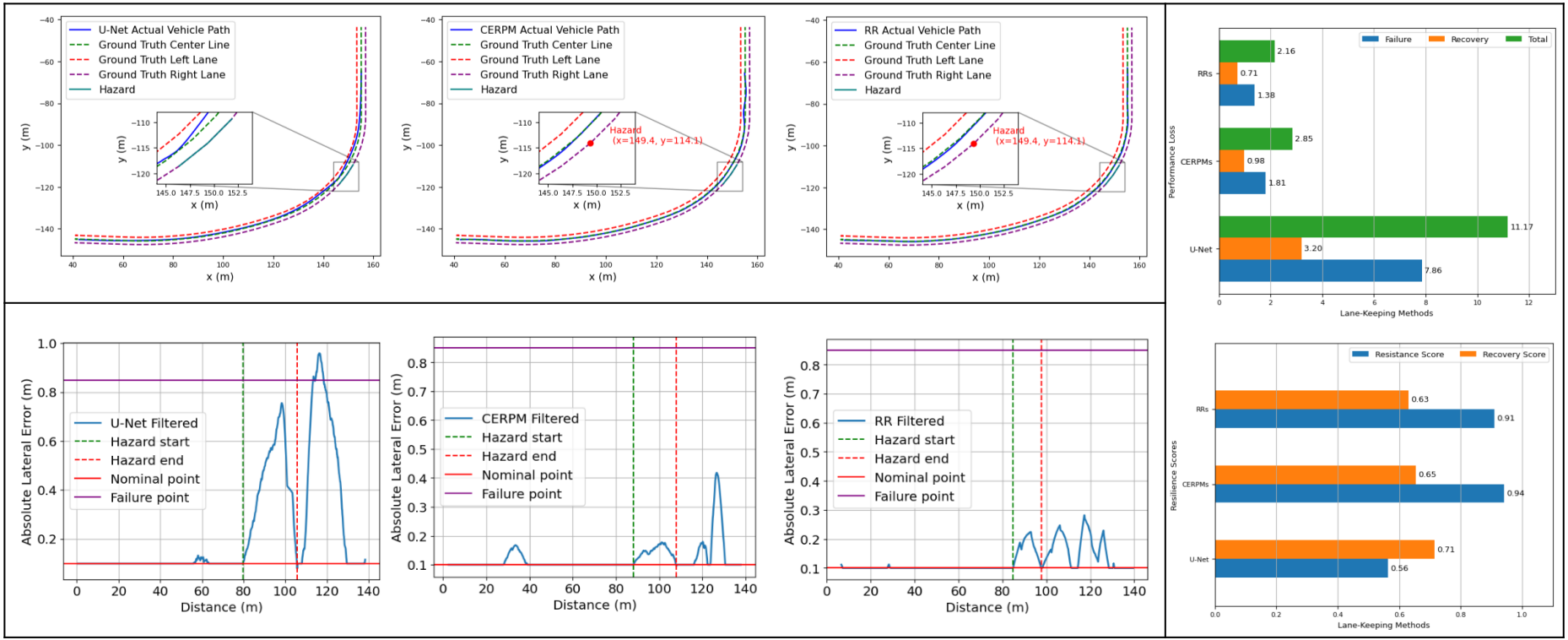


Late fusion based 3D object detection methods [Mao et. al.]

"AEPF: Attention-Enabled Point Fusion for 3D Object Detection" (2024)
Sharma et. al. Sensors

Technical Accomplishments & Progress

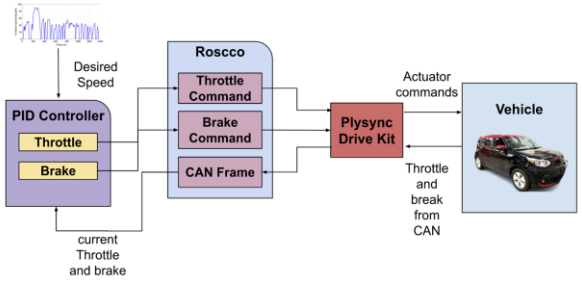
Proof That Vehicle Functionality is Maintained: Lane Line Detection Case Study

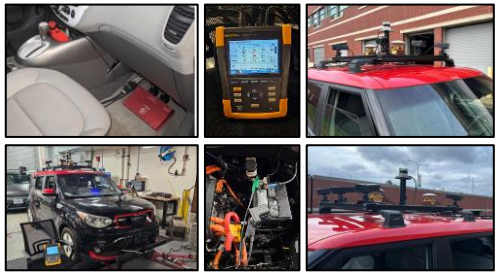


"Assessing Resilience in Lane Detection Methods: Infrastructure-Based Sensors and Traditional Approaches for Autonomous Vehicles" (2024)
Patil, Kadav, Fanas, Sharma, Masterson, Brown, Wang, Liao, Ekti, Asher. SAE World Congress

Technical Accomplishments & Progress

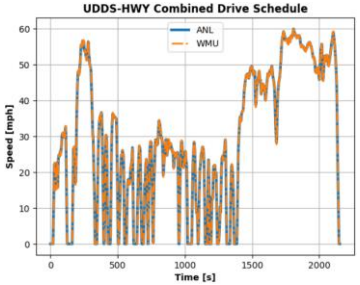
Energy Measurements

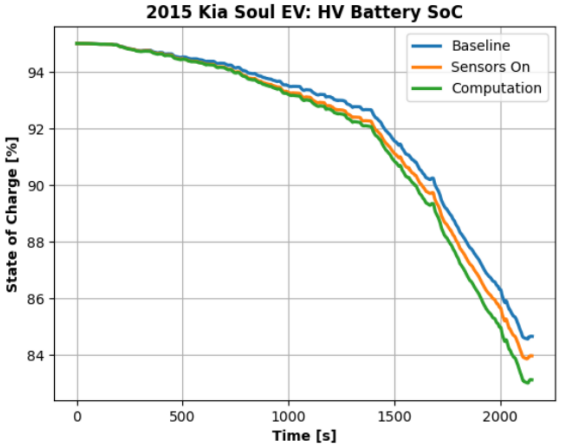




Validated against Argonne National Lab's Digital Dynamometer Database

➔ 1.64% difference in final state of charge

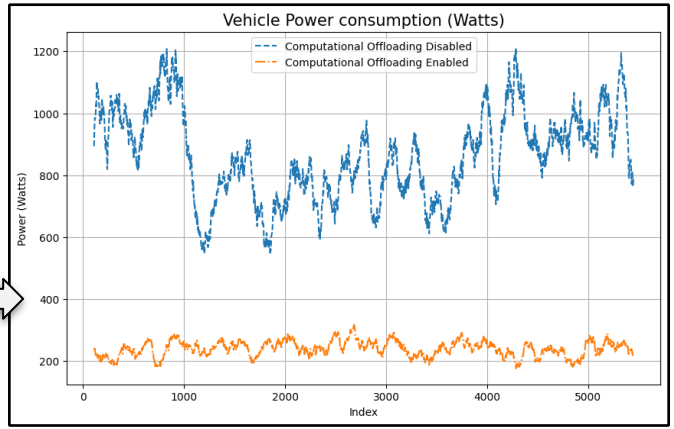




Δ SoC [%]	Range [mi.]
10.36	127.5
11.05 (7% ↑)	120.4 (5.6% ↓)
11.9 (15% ↑)	112.0 (12.2% ↓)

Chassis dyno data shows significant reduction in energy consumption from our infrastructure sensors

Academic publication in progress



"Evaluation of Autonomous Vehicle Sensing and Compute Load on a Chassis Dynamometer" (2023)
Brown, Motallebiaraghi, Fanas, Ayantayo, Meyer, Ekti, Wang, Goberville, Feinberg, Asher. IEEE Intelligent Transportation Systems Conference (ITSC)

Stakeholder Engagement

Organization	Names and Titles of Attendees	
City of Portage, MI	Muhammad Arif	Traffic Engineer
	Kendra Gwin	Director of Transportation & Utilities
	Eric Feldt	City Planner
	Devin Mackinder	Technology Services Director
MI Future Mobility & Electrification (OFME)	Kathryn Snorrason	Vice Pres. and Deputy Chief Mobility Officer
City of Kalamazoo, MI	Dennis Randolph	Public Works Manager
Metro	Sean McBride	Executive Director
	Kathy Schultz	Transit Planner
Grand Valley Metropolitan Council (GVMC)	Laurel Joseph	Director of Transportation Planning
Montana Dept. of Transportation (MDT)	Chris Dorrington	Director of Transportation
	Dustin Rouse	Chief Engineer
	Dwane Kailey	Chief Operating Officer
Utah Dept. of Transportation (UDOT)	Blaine Leonard	Transportation Technology Engineer
	Brett Slater	Program Manager
	Phillip R. Castro	Project Manager
	Troy L. Peterson	Director of Operations
Oregon Dept. of Transportation (ODOT)	Blaine Van Dyke	ITS Electrical Engineer
	Jason P Shaddix	ITS Manager
Colorado Dept. of Transportation (CDOT)	Michael Chapman	Winter Ops Program Manager
Kalamazoo Area Transportation Study (KATS)	Steve Stepek	Project Manager
Michigan Dept. of Transportation (MDOT)	Michele Mueller	Manager of Connect. & Auto. Veh. & Electrification
Federal Highway Administration (FHWA)	David Johnson	Lead of Road Weather and Work Zone Management

Collaboration and Coordination with Other Institutions

OAK RIDGE National Laboratory	**Oak Ridge National Laboratory** (national lab subaward) - Development and evaluation of Chip-Enabled Raised Pavement Markers (CERPMs) and Radar Retroreflectors (RRs). Assistance with vehicle integration and testing. Project students have interned here during summers.
NCAR	**National Center for Atmospheric Research** (national lab subaward) - Evaluation and characterization of weather data and its impact on weather sensor use.
DriveU.auto	**DriveU.auto** (industry subaward) - Development and evaluation of cellular network computational offloading using their existing cellular network teleoperational technology that is currently deployed in autonomous vehicle pilots worldwide. Assistance with vehicle integration and testing.
Revision Autonomy	**Revision Autonomy** (stakeholder) - Commercialization of CERPM technology by creating a business model and minimum viable product to be sold to infrastructure managers.

- Recurring meetings: monthly team-wide meetings, bi-weekly 1-on-1 meetings
- Identified relevant work at Argonne National Lab and Sandia National Lab, we meet as needed

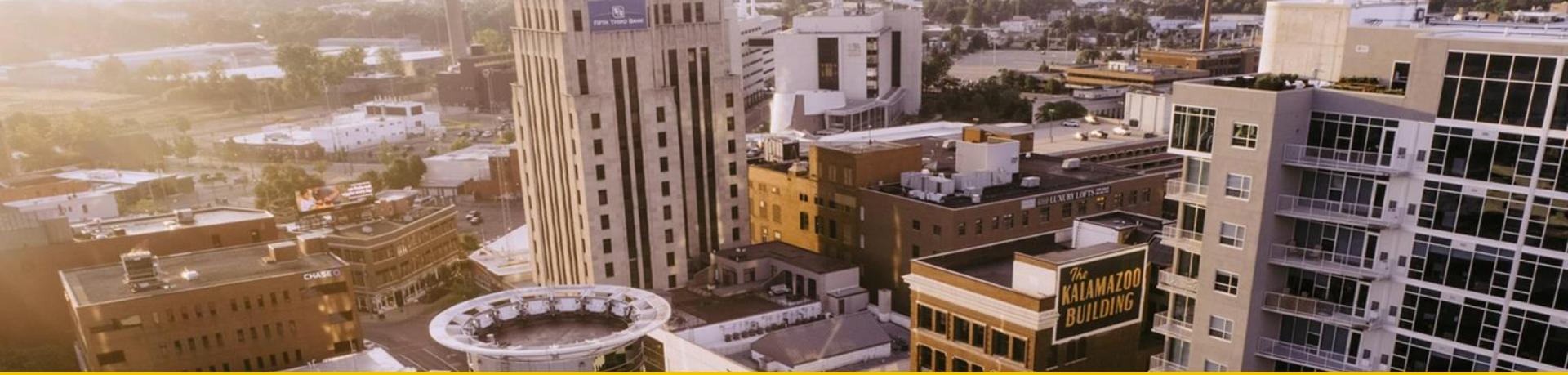
Summary

<u>Automation technology creates additional energy use and reduces the range of electric vehicles</u>

Key Takeaway: Infrastructure-based reduce this energy consumption & improve operational safety

Phase	Status
Year 1: Initial Development and Viability Demonstration for All Sensors	Complete
Year 2: Vehicle Integration of All Sensors with a 2015 Kia Niro Electric Vehicle	Complete
Year 3: On-Road Demonstration of Along Fixed Routes in Michigan and Tennessee	Complete

Key technologies from this project are currently being commercialized to maximize impact



Reviewer-Only Slides

Publications and Presentations

1. Sharma, Sachin, Johan Fanas Rojas, Ali Riza Ekti, Chieh Ross Wang, Zachary Asher, and Rick Meyer. 2023. “Vehicle Lateral Offset Estimation Using Infrastructure Information for Reduced Compute Load.” SAE Technical Paper.
2. Sharma, Sachin, Ali Riza Ekti, Johan F. Rojas, Nicolas E. Brown, David Pesin, Chieh Ross Wang, Shean Huff, Tim J. LaClair, Zachary D. Asher, and Richard T. Meyer. 2022. “Development and Evaluation of Chip-Enabled Raised Pavement Markers for Lane Line Detection.” In 2022 IEEE Sensors, 1–4.
3. Walker, Curtis Louis, Amanda R. Siems-Anderson, Zachary D. Asher, Parth Kadav, Farhang Motallebiaraghi, Nicholas Eric Brown, and Johan Fanas Rojas. 2023. “Energy Efficient Autonomous Vehicle Operation During Various Weather Conditions.” In 103rd AMS Annual Meeting. AMS.
4. “Automated Lane Centering: An Off-the-Shelf Computer Vision Product vs Infrastructure-based Chip-Enabled Raised Pavement Markers” (In Review) Kadav, Sharma, Fanas, Patil, Wang, Ekti, Meyer, Asher. MDPI Sensors
5. “Real World Use Case Evaluation of Radar Retro-reflectors for Autonomous Vehicle Lane Detection Applications” (2024) Brown, Patil, Kadav, Sharma, Fanas, Meyer, Hong, Liao, Ekti, Wang, Asher. SAE World Congress
6. “Engineering Requirements that Address Real World Hazards from Using High-Definition Maps, GNSS, and Weather Sensors in Autonomous Vehicles” (2024) Masterson, Patil, Brown, Fanas, Anderson, Walker, Rabinowitz, Asher. SAE World Congress
7. “Assessing Resilience in Lane Detection Methods: Infrastructure-Based Sensors and Traditional Approaches for Autonomous Vehicles” (2024) Patil, Kadav, Fanas, Sharma, Masterson, Brown, Wang, Liao, Ekti, Asher. SAE World Congress
8. “Evaluation of Autonomous Vehicle Sensing and Compute Load on a Chassis Dynamometer” (2023)
9. Brown, Motallebiaraghi, Fanas, Ayantayo, Meyer, Ekti, Wang, Goberville, Feinberg, Asher. IEEE Intelligent Transportation Systems Conference (ITSC)